Adaptive Multi-Regime Quantitative Trading System (AMR-QTS)

Master Plan v3.0 — Institutional Techniques at Retail Scale

> [!NOTE] v3.0 integrates all 6 hedge fund approaches from the Deep Critique — adapted for retail infrastructure (no Bloomberg, no CME feeds, no co-location). Changes from v2.0 are marked **[v3]**.

OBJECTIVE

Primary Goal: Generate consistent, compounding returns on personal capital through prop firm scaling.

Design Philosophy: - **[v3] Alpha-first** — prove the edge exists *before* engineering around it - Regime-adaptive via probabilistic models, not fixed thresholds - Risk-first architecture — survival before growth - Portfolio-level thinking, not isolated trade decisions - Fully systematic — zero discretionary interference - Statistically validated to institutional rigor standards

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ALPHA MODEL

Edge Hypotheses

#	Hypothesis	Factor	Free Data Source
1	**Momentum persistence** — trending regimes show continuation over 1–4 hour windows due to institutional order flow building over time	Momentum	MT5 OHLCV

#	Hypothesis	Factor	Free Data Source
2	**Mean reversion in constrained ranges** — 2σ deviation from mean with no directional flow reverts due to market-making behavior	Mean Reversion	MT5 OHLCV
3	**[v3] Volatility regime premium** — correctly classifying regime (not just trending/ranging, but the *probability* of being in that regime) adds alpha by avoiding false signals	Regime timing	MT5 OHLCV + DXY

Phase 1 Validation Requirements

Test	Tool	Pass Criteria
Random entry benchmark	Custom Python	Actual entries must outperform random by ≥ 15% on expectancy
Signal decay curve	Custom Python	Favorable excursion must increase for 10+ bars after entry
PnL attribution	Custom Python	Entry timing must contribute ≥ 20% of total PnL
Remove top 10% trades	Custom Python	System remains profitable
[v3] Naked signal test	Custom Python	Test each entry signal with no risk management — just entry → fixed hold → exit. If this is negative, the entry itself has no edge. Still useful as a risk-alpha system but changes how you think about the architecture.

INSTRUMENT SELECTION

Tier 1 — Core (Always Active)

Pair	Execution TF	Modules
EURUSD	15m	All
GBPUSD	15m	All

Tier 2 — After 3+ Months Validated

Pair	Execution TF	Modules
USDJPY	15m	Trend, High Vol
GBPJPY	15m	Trend, High Vol (50% position size)

Tier 3 — Optional

Pair	Execution TF	Modules
XAUUSD	15m	High Vol only

[v3] Cross-Asset Regime Signals (Free Data)

These are NOT traded — they are used as *inputs to the regime detector*:

Instrument	Source	What It Tells You
DXY (US Dollar Index)	TradingView free / MT5 symbol `USDX`	Dollar-wide trend vs range — if DXY is trending, all USD pairs are more likely in trend regime
VIX (or VIXM proxy)	Yahoo Finance / yfinance	Global risk appetite — VIX > 25 = risk-off, increases volatility regime probability
US10Y yield	FRED API (free) / yfinance	Rate differential driver — rapidly moving yields can cause regime breaks before FX price shows it

> These cross-asset signals feed into the HMM as additional observable features. They cost nothing but add early-warning capability that most retail systems lack.

Timeframe Stack

```
Daily      → Overall bias & key S/R
H4         → Regime confirmation, structural trend (HH/HL)
H1         → Regime classification (HMM + indicators)
15m        → Entry execution
```

Active Sessions

Session	Pairs	Modules
London Open (07:00–10:00 GMT)	EURUSD, GBPUSD	Breakout + trend
London/NY Overlap (12:00–16:00 GMT)	All Tier 1+2	All
Asian (00:00–07:00 GMT)	USDJPY	Mean reversion only

SYSTEM ARCHITECTURE — 5 LAYERS

> [!IMPORTANT] v3 adds **Layer E — Execution Layer** and restructures Layer B into a probabilistic model. The architecture is now 5 layers.

LAYER A — Risk Governance (Overrides All)

A1. Risk Per Trade

- Base risk: **0.5% equity**
- Max risk: **1%** — after equity ATH + 3 months consistency

A2. Daily Loss Cap

- Hard stop at **3% equity loss** → disabled for the day

A3. Open Exposure

- Max **3 simultaneous trades**

- Total exposure ≤ **2% equity**
- **[v3]** Total exposure is now calculated at the *portfolio level*, not per-trade — see Layer B2

A4. Correlation Control

- **Phase 1–4:** Max 1 USD-directional trade, max 1 GBP pair
- **[v3] Phase 5+:** Rolling 20-day correlation matrix. Second correlated trade allowed only if correlation < 0.7. If allowed, combined size reduced by correlation factor

A5. Volatility Safety Switch

- H1 ATR(14) > 2× 30-day SMA → reduce size 50%, disable mean reversion, tighten existing stops to 1× ATR

A6. News Filter

- No entries 30 min before / 15 min after high-impact events
- Existing trades: stops manage them

A7. Session Filter

- No entries in last 60 min before session close
- No trades Friday 20:00+ GMT / Sunday before 22:00 GMT

A8. Regime Transition Protocol

Transition	Action
TREND → RANGE/NEUTRAL	Tight ATR × 1.0 trail, exit if no progress in 8 H1 bars
RANGE → TREND	Exit mean reversion trades immediately
ANY → HIGH VOL	Tighten all stops to ATR × 1.0
HIGH VOL → ANY	Let existing trades run

LAYER B — Regime Detection Engine [v3 — REBUILT AS PROBABILISTIC MODEL]

[!IMPORTANT] **Hedge Fund Approach #2 fully implemented.** The regime detector is now a Hidden Markov Model (HMM) instead of fixed ADX thresholds. This is implementable for free using Python's `hmmlearn` library.

B1. Hidden Markov Model (HMM) Regime Detector

What is HMM in plain language: Instead of saying "ADX > 25 = trending," the HMM looks at *multiple features simultaneously* and says "given everything I see right now, there's a 78% probability we're in a trending state, 15% ranging, 7% high-vol." It learns the transition patterns — e.g., how often trends become ranges, how long each state lasts — from historical data.

Observable Features (inputs to HMM):

Feature	Calculation	Source
Returns	Log returns of H1 close	MT5 data
Realized volatility	Rolling 20-bar std dev of H1 returns	MT5 data
ADX(14)	Standard directional index, H1	MT5 data
ADX slope	ADX current − ADX 3 bars ago	MT5 data
ATR ratio	ATR(14) / ATR(14) 30-day SMA	MT5 data
EMA proximity	abs(EMA50 − EMA200) / ATR(14), H1	MT5 data
[v3] DXY trend	20-bar rate of change of DXY	MT5 / TradingView
[v3] VIX level	Current VIX (daily, carried forward to H1)	yfinance (free)

Hidden States (what the HMM classifies):

State	Description	Active Module
State 0	TRENDING LOW VOL	Module 1
State 1	RANGING LOW VOL	Module 2
State 2	HIGH VOLATILITY	Module 3

Decision Rules:

```
regime_probability = hmm.predict_proba(current_features)

IF max(regime_probability) > 0.70:
    → Activate corresponding module

IF max(regime_probability) < 0.70:
    → NEUTRAL — no new entries (high uncertainty)

Transition smoothing:
    → Require 2 consecutive H1 bars of >70% probability
      before switching regime (prevents whipsawing)
```

Training: - Train on 2+ years of H1 data in Phase 1 - Use Baum-Welch algorithm (built into hmmlearn) - Validate with walk-forward: retrain every 6 months on rolling data - 3 hidden states is the starting point — test 4 states (splitting trending into up/down) if 3 is too coarse

Fallback: If HMM is unreliable on live data, the v2 rule-based detector (ADX slope + ATR ratio) serves as the backup. Both run in parallel during Phase 4 demo — compare accuracy.

B2. Portfolio-Level Risk Budgeting [v3 — NEW]

> **Hedge Fund Approach #3 implemented.** Instead of "0.5% per trade" independently, allocate risk at the portfolio level.

Concept: The system has a total daily risk budget of **1.5% equity** (half the daily cap of 3%, providing margin of safety). This budget is *allocated* to strategy modules based on current regime confidence:

```
Total daily risk budget: 1.5%

Allocation by regime:
  IF TRENDING with >85% confidence:
    → Trend module gets 1.0% budget, Mean Rev gets 0.0%, High Vol gets 0.5%

  IF RANGING with >85% confidence:
    → Trend gets 0.0%, Mean Rev gets 1.0%, High Vol gets 0.5%

  IF HIGH VOL with >85% confidence:
```

```
    → Trend gets 0.0%, Mean Rev gets 0.0%, High Vol gets 1.0%
    (remaining 0.5% unallocated — safety buffer)

  IF MIXED confidence (<85% on any):
    → Split evenly: 0.5% / 0.5% / 0.5%
    → But NEUTRAL modules get 0.0%
```

Per-trade risk within budget: - Each module can take max **0.5% per trade** (unchanged) - But total module exposure capped by its allocated budget - Example: Trend module has 1.0% budget → can take 2 trades at 0.5% each

Why this matters: In simple per-trade risk, you might stack 3 trend trades at 0.5% each = 1.5% exposure to one regime. With portfolio budgeting, you consciously decide: "trending is high confidence right now, so it gets more of the risk budget." This is how hedge funds prevent concentration risk.

LAYER C — Strategy Modules

MODULE 1 — Trend Continuation

Activation: TRENDING regime with >70% HMM probability

HTF Confluence: - H4 structure trending (HH/HL or LH/LL) - Price above H1 EMA(200) for longs / below for shorts

Entry (15m): - Break and close above 20-bar high (longs) / below 20-bar low (shorts) - ATR expansion: current bar range > 1.5× avg of last 5 bars - RSI(14): 45–70 for longs, 30–55 for shorts - No entry within 1× H4 ATR of major H4 S/R level - **[v3]** Entry via limit order at breakout level (see Layer E)

Stop Loss: ATR(14) × 1.5 from entry

Exit: - Primary: 1:2 RR - ATR trailing activated at 1:1 - Excursion exit: retrace 50%+ from MFE before 1R → exit

Risk: 0.5% base (within Module 1's portfolio budget)

MODULE 2 — Mean Reversion

Activation: RANGING regime with >70% HMM probability

HTF Confluence: - H4 price inside established range - H1 EMA proximity < 0.5 (ATR-normalized)

Entry (15m): - RSI(14) < 30 → Long / RSI(14) > 70 → Short - Price at Bollinger Band extreme (2σ) - Confirmation: next candle closes back inside band - **[v3]** Entry via limit order at band level (see Layer E)

Stop Loss: ATR(14) × 1.2 beyond BB extreme

Exit: - Primary: Bollinger midline - Min RR: 1:1.5 - Time exit: 20 bars → exit at market

Risk: 0.5% base

Blocks: Disabled if ATR ratio > 1.5, if H4 trending, or if regime transitions to TRENDING.

MODULE 3 — High Volatility Protocol

Activation: HIGH VOL regime with >70% HMM probability

HTF Confluence: H4 directional break with momentum

Entry (15m): - 20-bar high/low breakout - Bar range (H−L) > 2× avg of last 20 bars - Candle body > 60% of total range - ATR current > previous ATR by 20%+ - **[v3]** Entry via limit order at breakout level (see Layer E)

Stop Loss: ATR(14) × 2.0

Exit: 1:1.5 RR minimum. No trailing stop.

Risk: 0.25% (mandatory 50% reduction)

LAYER D — Equity Curve & Performance Monitoring [v3 — MAJOR UPGRADE]

> **Hedge Fund Approach #6 fully implemented.** Replaces quarterly manual review with daily automated monitoring.

D1. Equity Curve Control (from v2)

- 20-period SMA of equity curve
- Equity < SMA(20) → reduce all sizes by 50%

- New equity ATH → restore base risk

D2. Automated Performance Monitor [v3 — NEW]

A Python script running daily (or a dashboard) that checks:

```
class PerformanceMonitor:
    """Runs daily after market close. Uses only trade log CSV."""

    def check_sharpe_deviation(self, realized_trades, backtested_sharpe):
        """
        Compare rolling 30-day realized Sharpe vs backtested.
        ALERT if realized < 50% of backtested for 15+ consecutive days.
        SHUTDOWN if this persists for 30+ days.
        """

    def check_regime_accuracy(self, predicted_regimes, actual_outcomes):
        """
        What % of regime predictions led to profitable trades?
        ALERT if accuracy drops below 55% over 30-day window.
        Signals HMM may need retraining.
        """

    def check_fill_deviation(self, expected_entries, actual_fills):
        """
        Track slippage: actual fill - expected entry.
        ALERT if avg slippage > 0.5 pips consistently.
        May indicate broker execution issues.
        """

    def check_module_contribution(self, trades_by_module):
        """
        Is each module contributing positive PnL?
        ALERT if any module is net negative over 60-day window.
        Consider disabling that module.
        """

    def check_signal_decay(self, recent_signals):
```

```
        """
        Has the edge decayed? Compare signal decay curve
        of last 100 trades vs backtest baseline.
        ALERT if mean favorable excursion at bar 10 dropped > 30%.
        """
```

Alert Levels:

Level	Condition	Action
🟡 WARNING	1 metric triggered	Log, investigate manually, no system change
🟠 ALERT	2+ metrics triggered	Reduce risk by 50%, investigate urgently
🔴 SHUTDOWN	Sharpe deviation > 30 days OR 3+ metrics triggered	Pause system entirely, retrain HMM, re-validate in Python

Delivery: Telegram bot notification + CSV log. No paid services needed.

LAYER E — Execution Layer [v3 — ENTIRELY NEW]

> **Hedge Fund Approach #4 implemented at retail scale.** Replaces "market order on signal" with structured execution.

E1. Order Type Selection

Module	Order Type	Why
Module 1 (Trend)	**Buy/Sell Stop** at breakout level	Places order at the 20-bar high/low *before* the break happens. Fills on the break itself rather than chasing a close. Reduces slippage.
Module 2 (Mean Rev)	**Buy/Sell Limit** at Bollinger Band level	If price is approaching the band, place limit at the band. Gets better fill than waiting for close + market order.
Module 3 (High Vol)	**Buy/Sell Stop** at breakout level	Same as Module 1 but with wider stop gap for volatility

E2. Order Management Rules

```
1. Pending order expires after 4 H1 bars (4 hours) if not filled
   → prevents stale orders filling in wrong regime

2. If regime changes before fill → cancel pending order immediately

3. Maximum slippage tolerance: 1.0 pip
   → if fill deviates > 1.0 pip from order price, exit immediately
   (indicates abnormal conditions — news, gap, liquidity void)

4. No more than 2 pending orders at any time per pair
```

E3. Slippage Modeling (Phase 1 Python)

```
def apply_realistic_execution(signal, spread, slippage_model):
    """
    In backtest, don't assume perfect fills.

    Model:
    - Entry: signal price + spread + random(0, 0.3 pips)
    - Stop hit: stop price + random(0, 0.5 pips) adverse
    - Target hit: target price - random(0, 0.2 pips) adverse

    This reduces backtest fantasy by ~5-15% which is realistic.
    """
```

DEVELOPMENT ROADMAP — 5 PHASES

PHASE 1 — Research & Backtesting in Python (8–10 Weeks)

Duration increased from 6–8 to 8–10 weeks to account for HMM development and institutional-grade validation.

Step 1: Data Collection

```
# MT5 Python API for OHLCV — EURUSD, GBPUSD (minimum)
# Min 3 years of H1, 15m, 5m data
# Cross-asset: DXY (MT5 or TradingView), VIX (yfinance)
# Clean: remove zero-vol bars, holiday gaps, data errors
# Add: realistic spread + slippage model from Layer E3
```

Step 2: Build HMM Regime Detector

```
from hmmlearn.hmm import GaussianHMM

# 1. Engineer features from H1 data:
#    returns, realized_vol, adx, adx_slope, atr_ratio,
#    ema_proximity, dxy_roc, vix_level

# 2. Fit HMM with 3 states on training data
hmm = GaussianHMM(n_components=3, covariance_type='full', n_iter=200)
hmm.fit(training_features)

# 3. Label states by inspecting means:
#    - State with highest return variance = HIGH_VOL
#    - State with highest mean abs return = TRENDING
#    - State with lowest mean abs return = RANGING

# 4. Validate: walk-forward retrain every 6 months
# 5. Compare accuracy vs rule-based detector (v2) on same data
```

Step 3: Build Strategy Modules

```
def trend_strategy(data_15m, data_h1, regime_prob) -> list[Signal]:
    """Requires regime_prob['trending'] > 0.70"""

def mean_reversion_strategy(data_15m, data_h1, regime_prob) -> list[Signal]:
    """Requires regime_prob['ranging'] > 0.70"""
```

```
def high_vol_strategy(data_15m, data_h4, regime_prob) -> list[Signal]:
    """Requires regime_prob['high_vol'] > 0.70"""

def portfolio_risk_manager(signals, equity, budget_allocation,
                           correlations, daily_loss) -> list[SizedOrder]:
    """
    Portfolio-level sizing:
    1. Check total budget remaining for each module
    2. Check correlation with existing positions
    3. Apply execution model (stop/limit order pricing)
    4. Return sized orders or empty list if blocked
    """
```

Step 4: Isolate Backtest Each Module

Metric	Pass Threshold
Profit Factor	> 1.3
Max Drawdown	< 15%
Win Rate	> 40%
Sharpe Ratio	> 0.8
Expectancy	> 0 per trade
Stability	Profitable across all 3 years independently

Step 5: Edge Validation

Test	Pass Criteria
Random entry benchmark	Actual entries beat random by ≥ 15% on expectancy
Signal decay curve	MFE increasing for 10+ bars
PnL attribution	Entry timing ≥ 20% of total PnL
Remove top 10% trades	Still profitable

Test	Pass Criteria
Naked signal test	Entry → fixed hold → exit shows directional bias

[v3] Step 6: HMM vs Rules Comparison

Run both regime detectors on the same data:

Metric	Rules (v2)	HMM (v3)
Regime accuracy (% of trades profitable in detected regime)	?	?
False regime rate (trades taken in wrong regime)	?	?
System Sharpe using this detector	?	?
Regime change lag (bars late vs actual structural change)	?	?

> **Decision rule:** If HMM outperforms rules by < 10% on all metrics, keep the simpler rule-based detector. Complexity must earn its place. If HMM outperforms by ≥ 10%, adopt HMM as primary with rules as fallback.

PHASE 2 — Combined System + Institutional Validation (4–6 Weeks)

> Duration increased to accommodate CPCV and deflated Sharpe testing.

Tests (v2 retained)

1. Walk-Forward: 6-month train → 2-month forward → ≥ 70% windows profitable
2. Monte Carlo: 1,000 iterations, 95th percentile DD < 20%
3. Robustness: spread +50%, params ±20%, remove best 20 trades
4. Regime-conditional Sharpe: each regime must have positive Sharpe individually

[v3] Institutional-Grade Statistical Tests

> **Hedge Fund Approach #5 fully implemented.** All tools are free Python.

Test A: Combinatorial Purged Cross-Validation (CPCV)

```
# Instead of simple 70/30 split, use CPCV from Marcos López de Prado
# Library: none needed, implement from "Advances in Financial ML" Ch. 12
# Or use: mlfinlab (pip install mlfinlab, free/open source)

# How it works:
# - Split data into N groups (e.g., 6 bi-monthly groups across 3 years)
# - Test all combinations of train/test splits
# - Purge overlapping samples to prevent leakage
# - Result: distribution of Sharpe ratios, not a single number

# Pass criteria:
# - Median Sharpe across all CPCV folds > 0.5
# - No fold has Sharpe < -0.5 (no catastrophic failure scenario)
```

Test B: Deflated Sharpe Ratio

```
# Accounts for multiple testing bias: if you tested 10 strategy
# variations, your "best" Sharpe is inflated by selection bias.
#
# Formula (López de Prado, 2014):
# DSR = Prob(Sharpe > 0 | number of trials, skewness, kurtosis)
#
# Implementation: ~20 lines of Python (scipy.stats)
#
# Pass criteria:
# - DSR > 0.95 (95% confidence the Sharpe is genuinely positive)
# - Track N_trials honestly: every parameter you changed counts
```

Test C: Minimum Backtest Length

```
# Formula: MinBTL = (1 + (1 - skew * SR) + (kurtosis * SR²)/4)
#                   * (z_alpha / SR)² years
#
# Purpose: tells you if 3 years of data is even ENOUGH to validate
#          your strategy's Sharpe ratio statistically
#
```

```
# If MinBTL > 3 years → your backtest is too short to be trustworthy
# → need more data or a stronger signal
```

PHASE 3 — MT5 EA Architecture (4–6 Weeks)

File Structure

```
AMR-QTS/
├── MainEA.mq5
├── include/
│   ├── RegimeDetector.mqh     ← Rule-based (always runs as fallback)
│   ├── TrendModule.mqh
│   ├── MeanReversionModule.mqh
│   ├── HighVolModule.mqh
│   ├── RiskManager.mqh        ← Portfolio-level budgeting
│   ├── EquityCurveControl.mqh
│   ├── NewsFilter.mqh
│   ├── TransitionManager.mqh
│   └── ExecutionManager.mqh   ← [v3] Stop/limit order logic
├── python/
│   ├── hmm_model.pkl          ← [v3] Serialized trained HMM
│   ├── regime_server.py       ← [v3] Runs HMM, serves regime via socket/file
│   └── monitor.py             ← [v3] Daily performance monitor + alerts
└── logs/
    └── trades_log.csv
```

[v3] HMM Integration Pattern:

The HMM runs in Python (hmmlearn doesn't have an MQL5 equivalent). Two options:

Option	How	Pros	Cons
File bridge	Python writes regime + probability to a CSV every H1 bar. EA reads the file.	Simple, no networking	1-bar delay possible

Option	How	Pros	Cons
Socket bridge	Python runs a local TCP server. EA sends features, receives regime.	Real-time, no delay	More complex to build

Start with file bridge in Phase 4 (simpler). Upgrade to socket bridge in Phase 5 if latency matters.

Input Parameters

```
input bool   UseBreakout          = true;
input bool   UseMeanReversion     = true;
input bool   UseVolatilityFilter  = true;
input bool   UseEquityCurveCtrl   = true;
input bool   UseNewsFilter        = true;
input bool   UseHMMRegime         = true;    // [v3]
input bool   UseLimitExecution    = true;    // [v3]
input double BaseRiskPercent      = 0.5;
input double TotalDailyBudget     = 1.5;     // [v3] portfolio budget %
input int    MaxDailyLossPercent  = 3;
input int    MaxOpenTrades        = 3;
input double MaxSlippagePips      = 1.0;     // [v3]
input int    PendingOrderExpiry   = 4;       // [v3] hours
```

Logging

Every trade logs: timestamp, pair, module, regime state, **regime probability**, entry price, order type (stop/limit/market), expected vs actual fill, SL, TP, lot size, exit price, PnL, daily DD, ATR, ADX.

Alerting

Telegram bot: daily cap hit, equity ATH, regime change, Sharpe deviation, HMM confidence drop, fill slippage warning, EA error.

PHASE 4 — Demo Forward Test (3 Months Minimum)

Rules: Zero parameter changes. Zero manual trades.

Metric	Target
Equity curve slope	Positive
Max daily DD	< 3%
Max overall DD	< 8%
Monthly return	2–7%
Realized vs backtested Sharpe	Within 1σ
[v3] HMM vs rules accuracy	Track both, compare
**[v3] Avg fill slippage	< 0.3 pips EURUSD

[v3] Run both detectors in parallel: - HMM provides regime classification - Rule-based provides regime classification - EA trades based on the primary (HMM or rules — whichever won Phase 1 Step 6) - Log both predictions — compare accuracy at end of 3 months

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PHASE 5 — Prop Firm Deployment

Challenge risk: 0.3% base **No trades:** Friday 18:00+ GMT, 30 min around high-impact news

Scaling:

Stage	Risk	Condition
Survival	0.5%	3–6 months record
Scaling	0.75%	Consistent profitability + equity ATH
Diversification	0.75%	Tier 2 pairs, 2nd prop account, dynamic correlation matrix

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PERFORMANCE PROFILE (Realistic)

Metric	Target
Win Rate	45–58%

Metric	Target
Avg RR	1.5–2.0
Monthly Return	3–7%
Max Drawdown	5–8%
Profit Factor	> 1.3
[v3] DSR (Deflated Sharpe)	> 0.95

CORE PRINCIPLES

1. Risk management > strategy quality
2. Regime adaptation > single edge
3. Robustness > optimization
4. Survival > growth
5. Statistical validation > gut feeling
6. Know your alpha — if you can't explain why the trade makes money, don't take it
7. **[v3] Complexity must earn its place** — simpler model wins unless complex model outperforms by ≥ 10%

PYTHON LIBRARIES REQUIRED (ALL FREE)

Library	Purpose	Install
`MetaTrader5`	Data pull + EA bridge	`pip install MetaTrader5`
`hmmlearn`	Hidden Markov Model	`pip install hmmlearn`
`numpy`, `pandas`	Data manipulation	`pip install numpy pandas`
`scipy`	Deflated Sharpe, statistical tests	`pip install scipy`
`ta` or `ta-lib`	Technical indicators	`pip install ta`

Library	Purpose	Install
`yfinance`	VIX + US10Y data (free)	`pip install yfinance`
`mlfinlab` (optional)	CPCV implementation	`pip install mlfinlab`
`python-telegram-bot`	Alert notifications	`pip install python-telegram-bot`
`matplotlib`, `seaborn`	Charting	`pip install matplotlib seaborn`
`joblib`	HMM model serialization	Built into Python

Total cost: $0.

CHANGELOG v2.0 → v3.0

#	Hedge Fund Approach	What Changed	Section
1	Research alpha first	Naked signal test added, deeper edge validation	Alpha Model
2	Use regime models, not rules	Full HMM regime detector with 8 features including DXY/VIX	Layer B1
3	Portfolio construction > per-trade	Portfolio risk budgeting system with regime-based allocation	Layer B2
4	Execution as separate layer	Layer E added — stop/limit orders, slippage tolerance, pending order mgmt	Layer E
5	Extreme OOS rigor	CPCV, Deflated Sharpe, Minimum Backtest Length tests	Phase 2
6	Automated live monitoring	PerformanceMonitor class with 5 daily checks + 3-tier alert system	Layer D2

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